Staying Power: The Individual Effects of the H-1B Visa Lottery*

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Abstract

TODO

1 Introduction

Understanding how immigration and immigration policy affect U.S. labor markets has long been a central question for economists. Existing work predominantly focuses on how immigration affects U.S. firms and local labor markets, with a particular emphasis on native workers. A key element, however, that is overlooked by focusing only on aggregate effects, is how immigrant workers themselves are affected by immigration policy. Understanding how individual foreign workers are impacted by obtaining U.S. work authorization, what their counterfactual outcomes in the absence of the authorization might have been, and how these effects vary across different types of workers and over time, is crucial for understanding the efficiency and distributional consequences of immigration policy. Furthermore, accurately understanding how firms and workers currently utilize the immigration system, an important

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consideration for crafting effective immigration policy, requires tracking individual workers through the immigration process.

Estimating the individual-level effects of obtaining a work permit, however, is difficult for two reasons: first, because obtaining a work permit is generally not exogenous to worker outcomes, and second, because individual-level data on immigration status and visa receipt, when made available, are generally deidentified and therefore difficult to link to data on labor market outcomes. In this project, I overcome these two hurdles by using individual application data from the H-1B visa lottery. The H-1B visa is an employer-sponsored visa intended to help firms fill gaps in the domestic labor force by hiring highly-skilled foreign workers for specialized roles. For many years, demand for H-1B visas has far exceeded the annual cap of 85,000 visas, leading to a lottery system for visa allocation that creates random variation in visa receipt among applicants. Using probabilistic linkage methods, I link a subset of deidentified H-1B lottery applications between 2020 and 2023 to individual work histories from LinkedIn data provided by Revelio Labs. I am, to my knowledge, the first to construct such a panel of H-1B lottery applicants linked to individual-level outcomes.

Considerable past research has studied the effects of H-1B visa access on the U.S. labor market Bound et al. (2017); Doran et al. (2022); Mahajan et al. (2024). This literature, however, has entirely focused on firm and local labor market outcomes, and largely covers lotteries prior to 2010. Two key institutional details enable me to study individual-level effects for more recent years of the lottery. First, while U.S. Citizenship and Immigration Services (USCIS) keeps administrative records on H-1B lottery winners in all years, it was only starting with the 2020 lottery that USCIS began to also keep records of lottery *losers*. Thus, while most prior papers use a proxy for lottery applications to identify firm-level effects, the availability of data on the full universe of applicants allows me to directly observe both individual winners and losers.

The second key insight is that the composition of H-1B applicants has changed over time. The administrative data I use is deidentified and contains detailed information about the sponsoring employer but only basic demographic information about the applicant themselves. The canonical view of an H-1B applicant is that of a worker from abroad who is hired by the sponsoring firm if and only if they win the lottery and obtain an H-1B visa. In this setting,

it is essentially impossible to identify lottery losers: armed with only basic demographics and no employer information, the set of possible matches would be overwhelmingly large.

In reality, however, a large share of modern H-1B applicants are already working in the United States before the lottery, most commonly under F-1 Optional Practical Training (OPT).[1] For this group, the sponsoring firm often employs the worker before it enters the lottery. By subsetting to international students who were *already working for the sponsoring firm at the time of the lottery*, I can reduce the candidate pool from the entire LinkedIn universe to the much smaller set of individuals working at the sponsoring firm. Furthermore, because I select the candidate pool based only on information on employment prior to the lottery, this OPT subsample is not subject to selection bias based on the lottery outcome.

Using these two insights, I am able to construct a linked applicant panel that includes both lottery winners and losers who had already been working for the sponsoring firm before the lottery. I then estimate the causal effect of winning the H-1B lottery on the individual worker by comparing outcomes for winners and losers in this linked sample.[2] Identification relies on the standard lottery assumption that, conditional on ADE status, selection is independent of applicant potential outcomes. In my main specification, I also include firm-by-lottery-year fixed effects, which compare winners and losers within the same sponsoring firm and lottery year.

I find that winning the H-1B lottery increases individuals' probability of remaining in the United States by 0.53 percentage points ($0.6\%^{***}$) and of remaining at the sponsoring employer by 0.44 percentage points ($0.8\%^{**}$) two years after the lottery. Accordingly, winners are less likely to be observed in their home country (0.14 percentage points, $6.4\%^{*}$) and puzzlingly more likely to have enrolled in new education (0.42 percentage points, $1.2\%^{*}$), although these results flip in other samples. These effects are statistically significant but small in magnitude.

One likely reason for these small estimates is that my probabilistic linkage inevitably introduces measurement error, which attenuates treatment effects toward zero. I address

[1] The number of F-1 students obtaining H-1B visas grew from 29,000 in FY2008 to 52,000 in FY2024 .

[2] I condition only on Advanced Degree Exemption (ADE) status, an indicator for whether an individual received an advanced degree (Master's or higher) from a U.S. institution. ADE applicants face a different lottery process and generally experience higher rates of success in the H-1B lottery.

this issue in several ways by using alternate samples of higher quality matches. I find that the estimated effects are larger in those samples, consistent with the idea that measurement error is attenuating the estimates in the full sample, but the effects remain small even in the highest-quality subsamples.

Why might we see such small effects of the H-1B lottery on individual outcomes? The main consideration is that by design, I am subsetting to individuals who are already working for the sponsoring firm before the lottery, and therefore have an active work permit at the time of the lottery. If it is indeed the case that I am capturing individuals on OPT, as intended, I should see that the estimated effect of winning the lottery is larger for individuals who are closer to the end of their OPT runway. I find suggestive evidence that the effects of winning the lottery are stronger for applicants who are either one or three years post-graduation (the years when regular OPT and the STEM OPT extension respectively expire). This pattern suggests that the small average effects may reflect the fact that many lottery losers are able to remain at the firm or in the United States through OPT for some time after the lottery, and therefore the main effect of lottery loss may be to weaken firm attachment, reduce employment stability, or accelerate return migration only at the margin.

These findings speak to several literatures. First, to my knowledge I am the first to estimate the causal effect of winning the H-1B lottery on an individual worker. An extensive body of work studies the aggregate and firm-level consequences of the H-1B visa, including the impacts on native employment and wages (Kerr et al., 2015; Peri et al., 2015a; Doran et al., 2022; Mahajan et al., 2024) and innovation and productivity (Kerr and Lincoln, 2010; Peri et al., 2015b; Dimmock et al., 2022; Mayda et al., 2023). A related line of work emphasizes how immigration rules can shape firms' outside options and the global geography of work, for example by inducing offshoring or affecting destination choices outside the United States (Khanna and Morales, 2017; Brinatti and Guo, 2023; Glennon, 2024). Another literature highlights that temporary work visas may generate employer-specific rents by restricting worker mobility and tying legal status to the sponsoring employer (Kim and Pei, 2022). This paper complements those studies by shifting the unit of analysis from the firm to the worker. In particular, I am able to capture the responses of individual lottery losers, who are generally not visible in firm-level analyses. Additionally, by extending analysis to

more recent years of the lottery, I capture how the effects of the lottery have evolved over time as the composition of applicants has changed and as the institutional environment has evolved.

Second, I make a methodological contribution by developing a probabilistic linkage procedure tailored to matching deidentified administrative data to individual-level outcome data. Probabilistic record linkage has a long tradition in economics, tracing back to the foundational framework of Fellegi and Sunter (1969) and its early applications in economic history (Ferrie, 1996). The approach has since been applied widely to link individuals across historical census records (Abramitzky et al., 2014; Feigenbaum, 2016; Abramitzky et al., 2021; Bailey et al., 2020) and to merge large-scale administrative datasets (Enamorado et al., 2019). The central challenge in my application is more severe than in most: because the H-1B administrative data is fully deidentified and contains neither names nor unique identifiers, standard text-matching approaches are infeasible. My key methodological insight is to exploit the pre-lottery employment relationship as an identifying anchor. By restricting the candidate pool to individuals observed in the Revelio data as employed at the sponsoring firm *before* the lottery, I transform an otherwise intractable many-to-many matching problem—in which each application could plausibly correspond to any one of millions of users—into a tractable one, in which the candidate pool is limited to the much smaller set of individuals at the same firm. I then score candidate matches using probabilistic weights over available demographic fields (Enamorado et al., 2019). A well-known consequence of probabilistic linkage is that match error attenuates treatment effect estimates toward zero (Bailey et al., 2020; Abramitzky et al., 2021); I address this by constructing samples of varying match quality and showing that estimated effects increase monotonically with match confidence.

Finally, I study a subset of H-1B applicants who are likely to be international students with work permits through OPT. The economics literature on international students and U.S. labor markets establishes that F-1 enrollment is tightly linked to post-study work opportunities and that Optional Practical Training (OPT) serves as a key bridge from U.S. education into the skilled labor force, but it provides relatively limited evidence on downstream employer-level impacts. A first strand shows that access to U.S. work visas shapes both the quantity and quality of international student inflows (Kato and Sparber, 2013;

Shih, 2016). A second strand documents that OPT expansions increase retention and induce sorting into STEM fields and occupations, particularly among graduate students (Amuedo-Dorantes et al., 2019; Demirci, 2019; Amuedo-Dorantes et al., 2023). Related work shows that U.S. higher education functions as an important pathway into the skilled labor market, with a substantial share of foreign graduates initially entering through OPT and contributing to short-run skilled labor supply (Bound et al., 2015; Beine et al., 2023). A smaller literature examines labor-market impacts, finding heterogeneous effects across experience groups and sectors, including substitution for recent native graduates alongside complementarities with more experienced workers (Demirci, 2020; Amuedo-Dorantes and Furtado, 2019). However, most existing work operates at the country, field, or local labor-market level, leaving open how firms absorb OPT-driven labor supply and how these flows translate into firm outcomes.

This paper contributes by bringing the analysis to the individual level, studying how international students use the H-1B visa and OPT in practice, and how the H-1B lottery affects their labor market outcomes. My findings suggest that for this subset of applicants, the H-1B lottery may function less as a mechanism for initial entry and more as a mechanism for retention and longer-horizon attachment. That distinction matters for how one interprets both efficiency and distributional concerns. If many applicants are already present and productive in the United States before the lottery, then the main effect of lottery loss may be to weaken firm attachment, reduce employment stability, or accelerate return migration only at the margin. At the same time, the modest average effects should not be read as evidence that the lottery is unimportant. A policy that creates substantial uncertainty, repeated reapplication costs, and employer dependence can have large welfare consequences even when average reduced-form employment responses are numerically small.

The remainder of the paper proceeds as follows. Section **??** describes the institutional setting, with particular attention to the H-1B lottery and the role of OPT. Section 3 introduces the administrative H-1B records and the Revelio outcome data, and discusses my linkage approach and sample construction. Section 4 lays out the empirical design and organizes the results. Section 5 concludes.

2 Background and Context: The H-1B Visa Program

Since its inception in 1990, the H-1B visa has remained the primary pathway for temporary skilled workers to immigrate to the US. In this section, I describe key institutional background on the H-1B visa program, the lottery system used to allocate visas, and the context of the modern pre-H-1B worker and the role of Optional Practical Training (OPT).

2.1 The H-1B Visa

The H-1B visa is an employer-sponsored temporary work visa for workers in "specialty occupations," created by the Immigration Act of 1990 as a way for firms to fill skill shortages in the US labor market (U.S. Citizenship and Immigration Services, 2026).

There are three key institutional details of the H-1B worker visa that are important for this paper. First, the H-1B is employer-sponsored, meaning that employers must apply for the visa on behalf of the worker, and the worker's legal status is tied to the sponsoring employer. Workers cannot self-sponsor. For employers to sponsor an H-1B worker, they must first file a Labor Condition Application (LCA) with the Department of Labor, attesting that they will pay the worker the prevailing wage for the occupation and location, and that the employment of the H-1B worker will not adversely affect the working conditions of similarly employed US workers (U.S. Department of Labor, Employment and Training Administration, 2024).

Second, the H-1B is temporary, with a maximum duration in most cases of six years (three years of initial status plus one three-year extension) (U.S. Citizenship and Immigration Services, 2026). Finally, the H-1B is dual-intent, meaning that while it does not lead directly to permanent residency, it does not preclude workers from applying for permanent residency while on H-1B status. Many H-1B workers apply for employer-sponsored green cards under their H-1B employer, and while H-1B visas are portable across employers, pending green card applications are not, creating an incentive for workers to remain with their sponsoring employer until they receive their green card (Kim and Pei, 2022).

2.2 The H-1B Lottery

H-1B employers are categorized into two groups: 'cap-subject' and 'cap-exempt'. Cap-exempt employers, which include non-profits and higher-education institutions, do not face restrictions on the number of H-1B visas they can sponsor. Cap-subject employers, which include most for-profit firms, are subject to an annual cap on the number of H-1B visas they can sponsor. Throughout this paper I focus on cap-subject employers, which are the relevant population for the lottery-based identification strategy.

Since 2005, the number of cap-subject H-1B visas that can be allocated in any given year has remained capped at 85,000 (U.S. Citizenship and Immigration Services, 2021). This is partitioned into 65,000 visas for all applicants ('regular' cap) and an additional 20,000 visas reserved for applicants with an "Advanced Degree Exemption" ('ADE' cap), i.e. applicants with an advanced degree (Master's or higher) from a U.S. institution. In years where demand exceeds supply (as has been the case in every year since 2013), U.S. Citizenship and Immigration Services (USCIS) conducts a lottery to allocate the limited number of visas.[3]

Prior to 2020, firms were required to submit a full petition for each lottery applicant, including an approved LCA, an I-129 form with information about the specific worker listed on the application, and a $10,000 processing fee. Two separate lotteries, one for ADE applicants and one for regular applicants, were then conducted among all submitted petitions, and only the petitions of lottery winners were processed.[4] This meant that no information was retained on lottery losers and the cost of application was very high for employers. Once petitions of lottery winners were processed and approved, lottery winners could begin working no sooner than October of that year, six months after the lottery.[5]

Starting in 2020, USCIS implemented an electronic registration system for the H-1B lottery (U.S. Citizenship and Immigration Services, 2019). Under the new system, firms

[3]Technically, allocation of the H-1B visas is on a first-come, first-served basis: applications are accepted and processed on a rolling basis starting in April of each year, and once the cap is reached, no more applications are accepted. Since 2013, the cap has been reached on the first day of applications.

[4]The ADE lottery is conducted first among ADE-eligible applications. The regular lottery is subsequently conducted with all non-ADE applications **and** ADE applications not selected in the ADE lottery. ADE applications can therefore be selected through either of the ADE or non-ADE lotteries.

[5]The share of rejected petitions is very small, under 5% in most years.

first submit a short electronic registration with basic information on the lottery applicant (henceforth 'applicant') and a small processing fee of $10 during a registration window in March. USCIS then conducts the random lottery separately by ADE status and notifies selected applicants (henceforth lottery 'winners'). Lottery winners are then given a window of 90 days to submit a fully completed petition, including the LCA and $10,000 processing fee (U.S. Citizenship and Immigration Services, 2020). There were no changes to processing of petitions or the timeline for H-1B activation after the lottery. This new registration system substantially reduced the cost of application for employers and created an electronic record of lottery applicants, including lottery losers, the crucial source of data for this paper.

In reducing the cost of application, however, the new system also created scope for multiple registrations on behalf of the same beneficiary. During the years in my sample, the random selection operated at the registration level rather than at the beneficiary level, meaning that a worker with multiple registrations (through different employers) could receive more chances in the lottery.[6] Investigative reporting has documented that some employers, particularly staffing firms and outsourcing companies, have submitted large numbers of registrations on behalf of the same beneficiary through subsidiaries and shell companies, creating a lottery system that is not random at the beneficiary level (Mider et al.). I attempt to exclude these types of employers from the analysis, with further details on the data and exclusion criteria described in the data section.

2.3 Optional Practical Training (OPT) and H-1B Lottery Applicants

The issue at the heart of individual linkage between H-1B application data and any dataset of outcomes is that I have very limited information on lottery applicants who do not subsequently submit an H-1B petition. This issue is particularly stark if we view lottery applicants exclusively as individuals who are initially outside the United States, and work at the sponsoring employer if and only if they win the lottery. In this case, while we are able to link lottery winners more easily because we can limit our search to workers at the sponsoring

[6] USCIS later adopted a beneficiary-centric selection process in response to these integrity concerns, beginning after the years I study (U.S. Citizenship and Immigration Services, 2024a, 2025a).

employer, lottery losers are essentially impossible to link with only basic demographic data. My solution to this linkage problem will be to focus on a large and growing subset of lottery applicants *who are already working for the sponsoring employer at the time of the lottery.*

The central institutional principle behind my linkage strategy is the importance of Optional Practical Training (OPT). OPT is a flexible work authorization option available to F-1 (student) visa holders, intended to provide practical training opportunities for students. The job must be related to the student's field of study, but no employer sponsor is required and self-employment is allowed. Students on F-1 (student) visas can typically work for up to 12 months after completing their studies through post-completion OPT, and eligible STEM graduates can apply for a 24-month extension, yielding up to 36 months of work authorization in total (U.S. Immigration and Customs Enforcement, 2024; U.S. Citizenship and Immigration Services, 2024b). For many international graduates of U.S. universities, the typical sequence is therefore: graduate, begin working immediately for a U.S. employer under OPT, and then enter the H-1B lottery one or more times while already employed as a way to extend their work authorization and remain in the U.S. after their OPT expires.

While statistics on the exact number of H-1B applicants who are initially using OPT are limited, statistics published by USCIS show that in recent years, between 50 and 70% of H-1B petitions approved for initial employment were filed on behalf of beneficiaries already in the United States, and of those, between 70 and 80% were filed on behalf of beneficiaries with OPT status (U.S. Citizenship and Immigration Services, 2025b).[7] I estimate in a separate data source that 94% of F-1 visa holders who eventually transition to the H-1B visa use OPT between graduation and H-1B status.

By focusing on OPT users, I am able to restrict the relevant candidate pool for *all lottery applicants* to workers already observed at the sponsoring firm prior to the lottery date. This makes the linkage problem much more tractable and by using only pre-lottery linkage variables, allows me to obtain a linked sample that is not selected on post-lottery outcomes.

[7]Note that these numbers are not separately reported for cap-subject and cap-exempt H-1B petitions.

3 Data and Linkage

3.1 Administrative Data

My data on H-1B applications was obtained from U.S. Customs and Immigration Services (USCIS) via a Freedom of Information Act (FOIA) request from Bloomberg News (Bloomberg Graphics, 2025). It is a complete record of all cap-subject lottery applications in FY 2020-2023, including the applicant's country of origin, year of birth, gender, sponsoring employer, and lottery outcome. For lottery winners who subsequently filed a petition for an H-1B visa, I have much more detailed information including their college major, job title, previous visa status, etc. The data contain an indicator for whether H-1B applicants who filed a petition (i.e. lottery winners) were selected via the ADE lottery or the regular lottery, but since ADE lottery losers can still reenter the regular lottery, this is not equivalent to an indicator for ADE status and is also not available for lottery losers. I therefore impute ADE status on the Revelio side using education histories, as described below.

Duplicate Applicants The data include an indicator for whether an applicant is a 'duplicate' (i.e. whether the applicant was already named on another application in the same lottery year), but do not indicate the original application. Therefore while I can filter on this indicator to obtain a sample of applications that are unique at the applicant x year level, I am not able to fully exclude all applicants that submit multiple applications in the same year (since one of the applications will not be marked as a duplicate or indicated in any way). To address this as best as possible, I use the duplicate indicator to identify and exclude employer-year cells with *any* duplicate applications.

3.2 Revelio Data

My data on individual outcomes comes from LinkedIn profiles scraped and compiled by Revelio Labs. The data include individual profile details such as full name and current location, as well as detailed position and education histories. The position data include employer, start and end dates, location, job title, and imputed compensation for many positions. The education data include institution name, program start and end dates, degree,

and field.

3.3 Linkage Strategy and Sample Construction

I briefly outline the process I use to link the deidentified H-1B administrative records (henceforth 'H-1B' data) to the LinkedIn-based worker histories (henceforth 'Revelio' data). A full account of the linkage pipeline is provided in Appendix 6. The linkage proceeds in three stages. First, I build an employer crosswalk between the FOIA files and Revelio firms. Second, I impute the fields needed for person-level matching on the Revelio side. Finally, within each employer-year cell, I score candidate workers and carry forward the resulting match weights into the outcome analysis. Table 1 report summary statistics for the full sample of H-1B applications as well as at various stages of the linkage process.

Employer Crosswalk The first problem is to align employers in the H-1B data with employer identifiers in Revelio. Both datasets feature duplicate employers, alternative or incorrect spellings, messy location data, and limited other information. I use a variety of deterministic matching methods (including name similarity and overlap scoring, location similarities and overlap, firm size, and industry information when available) to generate a set of candidate matches for each employer in the H-1B data. I then use an open-source large language model (LLM) to evaluate and score the candidate matches. The resulting crosswalk is many-to-many at the firm-year level. I am able to match 93% of firm-years in the H-1B data to a Revelio firm. To make linkage more tractable, I further exclude firm-years with more than ten total H-1B applications or *any* duplicate applications (as defined above). The resulting sample contains 114,206 firm-years, covering 296,677 applications.

Revelio Imputation Because the H-1B data are deidentified, the person-level match cannot use names directly. Instead, I match Revelio users to H-1B applications based on a set of demographic fields available for all applicants in the H-1B data: country of birth, year of birth, and gender. I impute these fields on the Revelio side using a combination of education histories, name-based predictions, and career histories. I also use education histories to impute variables advanced-degree exemption (ADE) status, an important control in the

later analysis. I initally pull all Revelio users who report working in the U.S. at a sponsoring firm at any point starting in 2015, a pool of over 19 million users. After imputation, I exclude users from the linkage pool who report attending a U.S. postsecondary institution (eligible for OPT) but do not report attending high school in the United States (likely not immigrants).

Individual Linkage I begin the individual linkage by defining a candidate pool for each H-1B applicant. The candidate pool consists of Revelio users who I impute as plausibly being of the country of origin listed on the H-1B application, report having started work at the sponsoring firm up to three years prior to the lottery, and are still working at the firm at the time of the lottery.[8] I further restrict to candidates with relatively close sex and year of birth agreement. Within this candidate pool, I compute an applicant-candidate score that combines the exact degree of agreement on U.S. work history, country of origin, year of birth, and sex. I then convert these scores into within-applicant weights that sum to one across candidates, and compute all outcomes as a weighted sum of individual user outcomes. My preferred matched sample contains 167,598 applications linked to at least one candidate with a positive match score, probabilistically matched to 744,945 Revelio users (average of 4.4 candidates per application). In my preferred specification, which includes firm-year fixed effects, I necessarily exclude firm-years with only one application, leaving a sample of 70,471 applications.

3.4 Alternate Samples and Match Validation

I use several variants of the main sample to probe the sensitivity of results to match quality and sample composition. Figure 1 shows .... The baseline sample is the most inclusive, retaining all applications with at least one positive-score candidate.

'Pre-filt' Sample The 'pre-filtered' sample filters H-1B applications *before* matching to exclude applicants from Southern Asia, Mandarin-speaking East Asia, and anglophone countries. High application rates from India and China make it difficult to distinguish between

[8] Three years is the maximum allowable time graduates can work in the U.S. on OPT.

many similar candidates at the same firm, and make it difficult to identify non-Indian applicants from Southern Asia and non-Chinese applicants from Mandarin-speaking countries by name alone, while Anglophone countries are difficult to impute based on name alone.

One-to-One Sample The 'one-to-one' sample begins with the baseline sample and imposes a one-to-one matching restriction (within each lottery year) by implementing an algorithm that maximizes the joint match score across all applications.

Strict Sample(s) Strict samples impose post-hoc criteria on the match weights to retain only applications with high composite scores. Figure 1 displays the Q50 strict sample, which retains applications whose top candidate meets the criteria and has a composite score above the 50th percentile among all top candidates.

Match Quality Validation For lottery winners who subsequently file an H-1B petition, I am able to observe more details in the H-1B data. I subset my sample to lottery winners and compare the FOIA-observable characteristics of their petitions to the corresponding fields in their linked Revelio profiles. I can compare the highest education level attained at the time of application, the field of study listed, the job title (for a subset of petitioners), and the U.S. institution attended for applicants in the ADE lottery. Note that match rates are not expected to be perfect even for true matches, and even using fuzzy matching, since both FOIA and Revelio fields are noisy and may not correspond directly to each other (e.g. Computer Science may be written as CS, Comp Sci, or EE/CS). Table 2 reports weighted match rates across sample variants, showing that match quality is significantly higher for the one-to-one and strict samples than for the baseline or pre-filtered samples.

3.5 Outcomes

All outcomes are drawn from Revelio position and education histories and are measured as snapshots one, two, and three years after the lottery year (denoted with suffixes $t = 1, 2, 3$).[9] I organize outcomes into four groups, each forming an exhaustive and mutually exclusive

[9] Three-year outcomes are available only for lottery years 2020–2021 given data coverage.

Table 1: Summary Statistics: H-1B Applications by Lottery Outcome

	All H-1B		In-sample firms		U.S. Educ.	
Statistic	Winner	Loser	Winner	Loser	Winner	Loser
% Winner	31.7		37.4		40.0	
% ADE	23.9	—	31.8	—	34.7	—
% Female	32.0	30.4	41.1	39.0	41.3	40.0
Avg. Age at App.	31.1	32.4	30.5	31.6	29.4	30.1
% India	74.3	77.7	40.0	33.7	44.8	38.3
% China	10.5	6.6	20.5	16.2	21.8	17.9
N apps	1,804,147		296,677		167,598	
N firm × years	170,346		114,206		61,816	

% ADE is based on the FOIA field $S3Q1 = M$ and is shown only for selected applicants; loser cells are undefined by construction. N apps and N firm × years are sample-wide totals reported once per group; N firm × years counts distinct $foia_firm_uid$ × lottery-year cells. The U.S. Educ. sample is deduplicated to unique applicants using $foia_indiv_id$ before computing statistics.

Table 2: Ex-Post Match Quality Across Sample Variants

Dimension	FOIA cov.	U.S.-educ.	Pre-filtered	One-to-one	Strict Q50
Education level	26–33%	62.0%	61.7%	73.9%	85.7%
Institution (ADE)	—	34.4%	45.9%	46.6%	49.2%
Field exact	100%	19.6%	20.0%	27.9%	33.7%
Job title	100%	4.0%	3.7%	6.1%	9.0%
Ed. + any other	—	34.9%	34.6%	47.1%	60.3%
N (applications)		106,819	25,231	101,797	47,881

Notes: Weighted match rate = weighted mean of a binary agreement indicator using *weight_norm* as weights. FOIA cov. reports the share of applications in the sample with a non-missing FOIA validation field for that row. *Education level*: agreement on degree type (doctorate / master's / bachelor's / other). *Institution*: Levenshtein + Jaro-Winkler fuzzy match (≥0.85) on institution name, restricted to ADE applicants. *Field exact*: exact match on cleaned field-of-study strings. *Job title*: fuzzy match (≥0.75) between FOIA title and top-20 Revelio titles for that applicant. *Ed. + any other*: education-level match and at least one of institution, field exact, or job title agrees. Pre-filtered excludes Southern Asia, East Asia, UK, Canada, Australia. One-to-one uses optimal bipartite matching. Strict Q50 restricts to applicants whose top-candidate score exceeds the 50th percentile threshold.

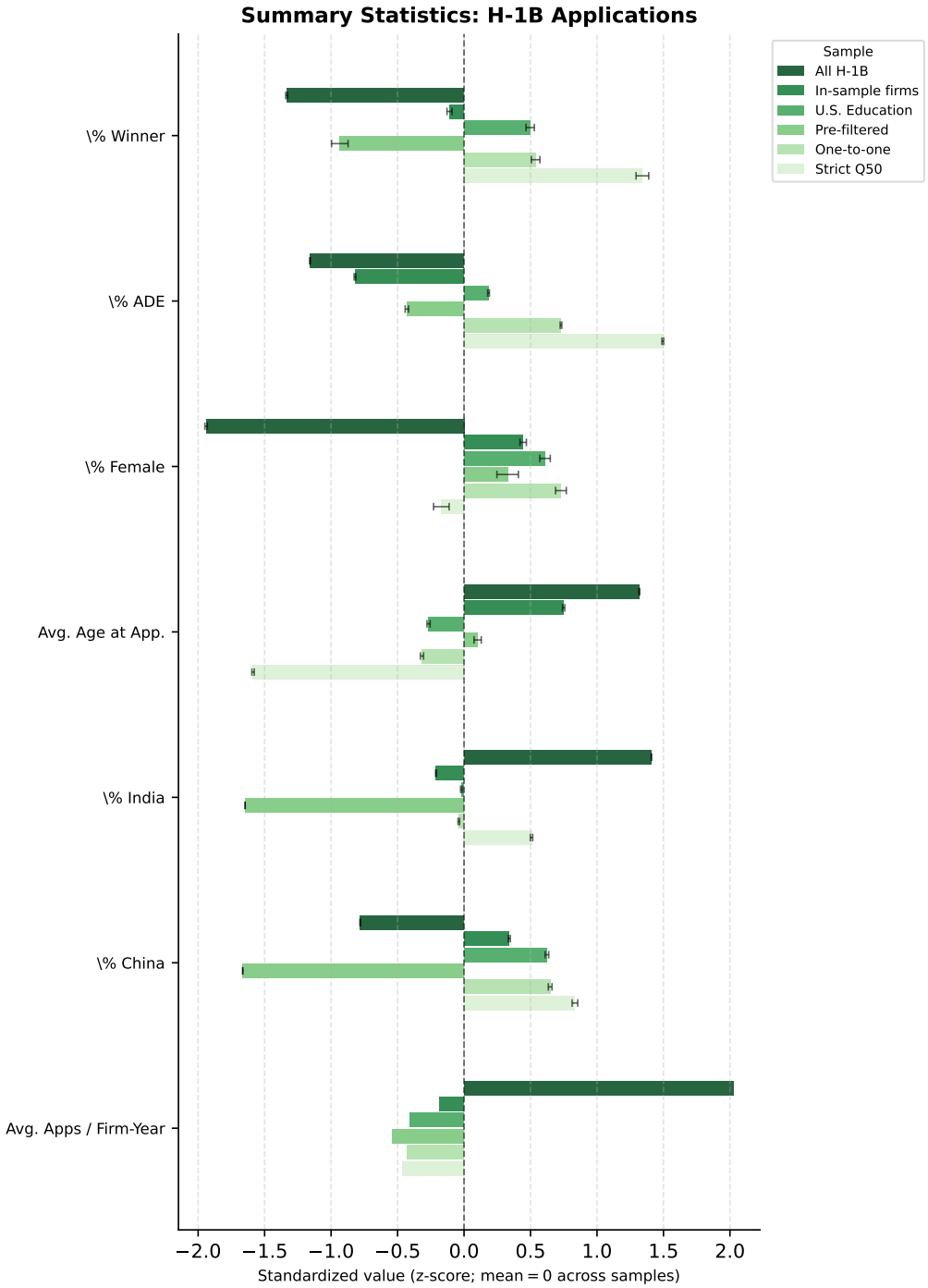

Figure 1: Summary Statistics: H-1B Applications. Each bar shows the z-score of that statistic for the given sample relative to the cross-sample mean and standard deviation (mean = 0, std = 1 across samples).

partition within its domain.

Work status. The primary employment outcomes partition each applicant's status at horizon t into three mutually exclusive states: *still at firm* (an active position at the sponsoring firm is observed), *different firm* (an active position at any other employer is observed, with no active position at the sponsoring firm), and *other* (no active position is observed at any employer). By construction, $still_at_firm_t + diff_firm_t + other_t = 1$. The *different firm* outcome is further decomposed into positions that started after the lottery (*new_diff_firm*) and positions that started before the lottery (*old_diff_firm*).

Within-firm mobility. Conditional on *still at firm* at horizon t, I decompose same-firm stayers into those who have started a new post-lottery position at the firm (*same_firm_new_position*, proxying for within-firm advancement) and those who remain in the same pre-lottery position. The latter is further split by whether the position's end date is null (imputed as ongoing) or explicitly recorded as non-null. Regressions on within-firm outcomes use only the subsample with $still_at_firm = 1$ at the corresponding horizon.

Location. Location outcomes are constructed from the location fields in position and education histories. At each horizon t, a worker is classified as *in the U.S.* if any active position or education spell lists a U.S. location, *in home country* if the highest-priority active record lists the applicant's country of birth, and *non-U.S. non-home* if an active record in a third country is observed. Workers with no active records or no location information fall into a fourth residual category (*location null*); these four states form an exhaustive partition.

Education. Education outcomes distinguish between *new education* (an active education spell whose start date falls after the lottery year) and *continuing education* (an active spell that began before the lottery). Workers with no active education spell are the residual category. Together with the residual, these two indicators form an exhaustive partition of education status at each horizon.

4 Results

4.1 Empirical specification

The main specification compares lottery-selected and non-selected applicants in the linked sample, conditioning on advanced-degree-exempt status because ADE applicants face a different lottery assignment process. For applicant i in lottery year t at firm c, I estimate:

$$y_{ict}^{s} = \beta_0 + \beta_1 Win_{ict} + \beta_2 ADE_{it} + \gamma_{ct} + \varepsilon_{ict}^{s}, \tag{1}$$

where y_{ict}^{s} indicates the outcome of interest measured s years after the lottery year t, Win_{ict} indicates lottery selection for applicant i in year t with firm c, ADE_{it} indicates eligibility for the advanced-degree exemption, and γ_{ct} denotes firm-by-lottery-year fixed effects. Outcomes are aggregated to the applicant level using the probabilistic match weights described in Section **??**, and standard errors are clustered at the firm level. The parameter β_1 captures the effect of winning the H-1B visa lottery on outcome y_{ict}. The key identifying assumption is that, conditional on ADE status and firm-lottery-year cells, lottery selection is as good as random and does not differentially affect the probability of appearing in the matched sample. The fixed effects control for any time-varying firm-level factors that could confound the relationship between lottery selection and outcomes.

My main outcomes of interest relate to individual location, employer switching, 'promotions', and education enrollment. For location, I measure whether the worker is observed in the United States or in their country of origin two years post-lottery. For employer switching, I measure whether the worker has an active position at the same firm or a different firm than the sponsoring firm two years post-lottery. For promotions, I measure whether the worker has started a new position at the same firm after the lottery, among applicants who subsequently remained at the same firm after the lottery. For education, I measure whether the worker has an active education spell started after the lottery. These outcomes capture key dimensions of labor market attachment and mobility that may be affected by lottery selection.

4.2 Main results

Table 3 presents the main regression estimates for six outcomes measured two years after the lottery, using the U.S.-educated linked sample (N = 70,471).[10] All regressions include firm-by-lottery-year fixed effects; standard errors are clustered by firm.

The point estimates for lottery winners are positive and statistically significant across several key outcomes. Winning the lottery raises the probability of being observed in the United States two years post-lottery by 0.53 percentage points (0.6%***) and reduces the probability of appearing in one's home country by 0.14 percentage points (6.4%**). On the employment margin, winners are 0.44 percentage points more likely to remain at the sponsoring firm (0.8%**) and equally less likely to hold a new position at a different employer (1.0%**), consistent with the visa reducing employer mobility. Winners are also marginally more likely to have enrolled in new education post-lottery (1.2%*). By contrast, the conditional probability of a new within-firm position—among applicants who remain at the same firm—shows no significant winner effect (0.1%, N = 8,350).

Table 3: Main Effects of H-1B Lottery Selection: Two Years Post-Lottery

	In U.S. (1)	In home ctry (2)	Same firm (3)	New pos. (cond.) (4)	New diff. firm (5)	New educ. (6)
winner	0.0053***	−0.0014**	0.0044**	−0.0006	−0.0044**	0.0042*
	(0.0012)	(0.0007)	(0.0021)	(0.0075)	(0.0021)	(0.0022)
ade	0.0053	0.0004	0.0125*	0.0008	−0.0125*	0.0017
	(0.0040)	(0.0019)	(0.0066)	(0.0220)	(0.0066)	(0.0067)
Control mean	0.911	0.022	0.543	0.444	0.457	0.348
N	70471	70471	70471	8350	70471	70471

Notes: OLS estimates. Sample: U.S.-educated linked sample (N = 70,471). All outcomes measured two years post-lottery. *In U.S.* = observed in the United States (from position or education location field). *In home ctry* = observed in country of origin. *Same firm* = indicator for active position at sponsoring firm. *New pos. (cond.)* = new post-lottery position at same firm, estimated on the subsample with *Same firm* = 1 (N = 8,350). *New diff. firm* = active position at a different firm started after lottery. *New educ.* = active education spell started after lottery. All regressions include firm-by-lottery-year fixed effects. Standard errors (in parentheses) clustered by firm. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

[10] Of 111,022 applications in the U.S.-educated linked sample, 40,551 are singletons—the only application in their firm-year cell—and are dropped because firm-year fixed effects are not identified for them.

4.3 Robustness to linkage quality and sample selection

Table 4 reports the winner coefficient for all six outcomes across five sample variants: the U.S.-educated baseline, the one-to-one sample (optimal bipartite matching that assigns each applicant to at most one Revelio candidate), the pre-filtered sample (excluding applicants from countries with high candidate multiplicity), and three strict-threshold samples that progressively restrict to higher-quality matches (Q25, Q50, and Q75 score thresholds). The key pattern is that winner effects grow monotonically with sample quality, consistent with attenuation from match error in noisier baseline cells.

In the pre-filtered sample (N = 10,425), winning increases two-year firm retention by 2.8 percentage points ($p < 0.01$) and reduces new employer switching by an equal amount ($p < 0.01$); the U.S.-presence effect is 1.0 percentage point ($p < 0.05$). In the strictest sample (Q75, N = 15,218), the U.S.-presence effect reaches 2.0 percentage points ($p < 0.01$), retention rises by 2.1 percentage points ($p < 0.05$), and the conditional within-firm new position effect becomes negative and significant (-3.0 pp, $p < 0.05$). These results support the interpretation that the baseline estimates, while already significant, are attenuated by measurement error and understate the true lottery effects for well-matched applicants.

The internal validity of the research design requires that lottery selection not differentially affect the probability of appearing in the matched sample. The external validity caveat is explicit: the paper speaks to U.S.-educated H-1B applicants already employed at sponsoring firms and visible in LinkedIn-based data before the lottery, rather than to all H-1B applicants worldwide.

4.4 Heterogeneity and Mechanisms

A particularly important heterogeneity dimension is the amount of time the worker appears to have remaining before OPT exhaustion. The code constructs a measure of time since graduation, which is a natural proxy for remaining OPT runway. The prediction is straightforward: effects of winning should be larger for workers who are closer to the end of their allowable OPT period. If the treatment effect grows when alternative authorization is scarce, that is exactly what the institutional context predicts.

Table 4: Robustness: Winner Effects Across Sample Variants

	(1) In U.S.	(2) In home ctry	(3) Same firm	(4) New pos. (cond.)	(5) New diff. firm	(6) New educ.
Panel: U.S.-educated (N = 70,471)						
Winner	0.0053***	−0.0014**	0.0044**	−0.0006	−0.0044**	0.0042*
	(0.0012)	(0.0007)	(0.0021)	(0.0075)	(0.0021)	(0.0022)
Ctrl. mean	0.911	0.022	0.543	0.444	0.457	0.348
Panel: One-to-one (optimal bipartite matching) (N = 64,712)						
Winner	0.0180***	−0.0072***	0.0160**	−0.0051	−0.0160**	0.0034
	(0.0039)	(0.0022)	(0.0065)	(0.0097)	(0.0065)	(0.0063)
Ctrl. mean	0.900	0.033	0.544	0.510	0.456	0.387
Panel: Pre-filtered (excl. Southern Asia, East Asia, UK, CA, AU) (N = 10,425)						
Winner	0.0103**	−0.0002	0.0281***	−0.0331*	−0.0281***	−0.0147*
	(0.0048)	(0.0026)	(0.0079)	(0.0186)	(0.0079)	(0.0079)
Ctrl. mean	0.895	0.027	0.556	0.476	0.444	0.345
Panel: Strict (Q25 score threshold) (N = 39,506)						
Winner	0.0142***	−0.0054**	0.0071	−0.0038	−0.0071	−0.0040
	(0.0038)	(0.0026)	(0.0059)	(0.0081)	(0.0059)	(0.0061)
Ctrl. mean	0.880	0.052	0.542	0.524	0.458	0.507
Panel: Strict (Q50 score threshold) (N = 28,325)						
Winner	0.0174***	−0.0080**	0.0074	−0.0163*	−0.0074	0.0011
	(0.0047)	(0.0033)	(0.0073)	(0.0098)	(0.0073)	(0.0077)
Ctrl. mean	0.873	0.061	0.539	0.536	0.461	0.521
Panel: Strict (Q75 score threshold) (N = 15,218)						
Winner	0.0196***	−0.0066	0.0205**	−0.0297**	−0.0205**	0.0074
	(0.0069)	(0.0055)	(0.0104)	(0.0134)	(0.0104)	(0.0108)
Ctrl. mean	0.877	0.062	0.545	0.553	0.455	0.524

Notes: Each cell reports the OLS estimate of $\hat{\beta}_1$ (winner effect) from equation (1) with standard errors clustered by firm in parentheses. All regressions include firm-by-lottery-year fixed effects. Outcomes measured two years post-lottery; see Table 3 for definitions. One-to-one uses optimal bipartite matching (Jonker–Volgenant) to assign each applicant to at most one Revelio candidate. Pre-filtered excludes applicants from countries with high candidate multiplicity (Southern Asia, East Asia, UK, Canada, Australia). Strict (Q25/Q50/Q75) restricts to applicants whose match quality score exceeds the 25th, 50th, or 75th percentile threshold, respectively. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5 presents heterogeneous effects on two-year same-firm retention (*Same firm*) across two dimensions: years since graduation at the time of the lottery (columns 1–4) and employer reputation class (columns 5–6). The graduation-year subgroups partition applicants by Δg, defined as lottery year minus one minus last graduation year; $\Delta g = 0$ and $\Delta g = 1$ correspond to workers still within their initial two-year OPT window, while $\Delta g \geq 2$ covers those on STEM-OPT extensions or other bridging arrangements. The reputation subgroups split by whether the sponsoring firm has no recorded reputation score (*norep*) or a high reputation score (*highrep*).

Table 5: Heterogeneous Effects on Two-Year U.S. Retention by Years Since Graduation and Employer Reputation

	graddiff= 0	graddiff= 1	graddiff= 2	graddiff= 3	no_rep	high_rep
	(1)	(2)	(3)	(4)	(5)	(6)
Panel: U.S.-educated (N = 4,534)						
Winner	−0.0002	0.0032	0.0006	0.0060	0.0099	0.0021
	(0.0027)	(0.0021)	(0.0023)	(0.0040)	(0.0075)	(0.0020)
Ctrl. mean	0.907	0.903	0.905	0.903	0.901	0.926
Panel: One-to-one (N = 1,985)						
Winner	0.0181	0.0365***	0.0334**	0.0390*	0.0197	0.0074
	(0.0208)	(0.0141)	(0.0169)	(0.0199)	(0.0222)	(0.0066)
Ctrl. mean	0.892	0.879	0.870	0.874	0.889	0.919
Panel: Strict (Q75) (N = 1,058)						
Winner	0.0116	0.0116	0.0036	0.0346**	0.1099	−0.0026
	(0.0186)	(0.0113)	(0.0120)	(0.0148)	(0.0721)	(0.0108)
Ctrl. mean	0.905	0.878	0.857	0.827	0.809	0.912

Notes: OLS estimates of $\hat{\beta}_1$ (winner effect) from equation (1). Outcome: *In U.S.* (active U.S. position two years post-lottery). Each panel corresponds to a sample variant; see Table 4 for definitions. One-to-one uses optimal bipartite matching to assign each applicant to at most one Revelio candidate. Strict (Q75) restricts to applicants whose match quality score exceeds the 75th percentile threshold. Within each panel, columns (1)–(4) restrict to applicants with Δg equal to the indicated value (rounded), where Δg = lottery year − 1 − last graduation year; columns (5)–(6) restrict to applicants at no-reputation and high-reputation employers, respectively. All regressions include firm-by-lottery-year fixed effects. Standard errors (in parentheses) clustered by firm. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.5 Interpreting the magnitudes

The central finding is that winning the H-1B lottery significantly increases U.S. presence and retention at the sponsoring firm while reducing new employer mobility. Two features of the results merit emphasis.

First, the baseline effects are likely attenuated by probabilistic linkage error. Even if the true individual-level effect is larger, mixing correct and incorrect candidate links in the weighted average compresses estimates toward zero. The monotone increase in estimated effects as match quality improves—from 0.53 pp (In U.S.) in the baseline to 2.0 pp in the Q75 strict sample—is consistent with this interpretation. The pre-filtered winner effect on two-year same-firm retention (2.8 pp) is more than six times the baseline point estimate (0.44 pp), suggesting the baseline substantially understates the true effect.

Second, even the baseline effects are economically meaningful for a subpopulation of U.S.-educated workers whose visa status is genuinely at stake. Among U.S.-trained workers already employed under OPT, losing the lottery raises separation risk and the probability of leaving the United States, but many losers find ways to remain through STEM-OPT extensions, cap-exempt employment, or repeat lottery applications. The H-1B is thus best characterized as a *stability lottery* rather than an entry lottery for this subgroup: winning reduces the risk of disruption and locks workers into the sponsoring firm, but does not transform a worker who would otherwise leave into one who stays. The largest effects appear for applicants with limited alternative pathways—those visible in higher-quality linked samples—for whom the lottery outcome is more binding.

5 Conclusion

This paper asks a simple but previously difficult question: what does winning the H-1B lottery do to the individual worker? I answer that question by combining recent deidentified H-1B administrative records with LinkedIn-based worker histories from Revelio Labs and by developing a probabilistic linkage strategy that can connect the two without direct personal identifiers. The approach is designed around the institutional reality that many recent H-1B applicants are already working for the sponsoring employer before the lottery, often under OPT. That fact makes the worker-level question empirically tractable.

The paper makes two main contributions. First, it introduces a linkage framework for deidentified immigration records that should be useful beyond this specific application. The contribution is not only that a link can be made, but that the uncertainty in the link

can be carried transparently into the analysis through match weights and high-precision subsamples. Second, it provides the first worker-level causal evidence, to my knowledge, on the consequences of winning the H-1B lottery. The current results indicate that lottery winners are more likely to remain with the sponsoring employer and more likely to remain in the United States, but the magnitudes are small.

Those small effects are themselves informative. They suggest that, for a large subset of recent applicants, the H-1B cap is not simply a gate that cleanly separates entry from non-entry. Instead, it interacts with a broader system of temporary work authorization in which many lottery losers continue to work, reapply, extend OPT, or transition through other channels. For the modern pre-H-1B worker, winning appears to increase labor-market stability more than it changes the extensive margin of being present in the United States at all.

This interpretation has two policy implications. The first is that reforms to the H-1B system should be evaluated not only in terms of how many workers they admit, but also in terms of how they change employer dependence, repeated application costs, and uncertainty for workers already trained and employed in the United States. The second is that the individual-level incidence of H-1B rationing may differ from the incidence suggested by firm-level studies: the main worker-level effects may operate through retention and mobility rather than through abrupt displacement.

There are several natural extensions. One is to study wage and career progression in more detail once the preferred compensation measures are finalized. Another is to connect the linked applicant panel to productivity outcomes, such as GitHub activity or entrepreneurial transitions. A third is to study return migration and third-country reallocation more directly. More broadly, the project shows that the deidentified nature of administrative immigration data, while a serious obstacle, need not prevent credible worker-level analysis when combined with modern linkage methods and a careful understanding of institutional context.

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6 Linking H-1B Applications to Revelio Profiles

This appendix gives a self-contained description of the linkage used to connect deidentified H-1B lottery applications to LinkedIn-based worker histories from Revelio. The central difficulty is that the administrative H-1B records contain no names or personal identifiers for the full applicant pool. A global search over all Revelio users would therefore be too noisy to be useful. The linkage becomes feasible because many recent cap-subject applicants are already working for the sponsoring employer when they enter the lottery, often under Optional Practical Training (OPT). I use that pre-lottery employment relationship as the anchor for the match.

The linkage proceeds in three steps. First, I align employers across the two datasets. Second, I construct on the Revelio side the limited set of demographic and education variables that can be compared to the administrative records. Third, within each sponsoring employer-year cell, I rank plausible worker profiles and carry the resulting match uncertainty forward into the analysis through applicant-level match weights rather than forcing a deterministic one-to-one match.

6.1 A. Employer crosswalk and in-scope employer-years

The first stage creates a crosswalk between employers in the H-1B files and employers in Revelio. This stage uses cleaned employer names, employer tax identifiers where available, location and industry information, and a stock of previously validated firm links. Straightforward cases are kept directly. Ambiguous cases are narrowed using fuzzy comparison rules and then reviewed with a large language model that determines whether a candidate Revelio company corresponds to the same economic employer. The model is used only for firm identity resolution, such as rebrandings, spelling variants, and subsidiary relationships; it is not used to identify individual workers.

The employer crosswalk is intentionally many-to-many at the employer-year level. A single economic employer can appear under more than one vendor-specific company identifier in Revelio, and collapsing those identifiers too early would create false non-matches downstream. The pipeline therefore carries forward all valid employer-year links and later uses

workers' own position histories to give more weight to the most credible employer mapping.

Before any person-level matching takes place, the code also defines a conservative employer-year sample. The current working sample keeps only employer-year cells with no duplicate registrations and at most ten H-1B applications. This restriction removes the highest-multiplicity cells, where individual matching is least reliable and where outsourcing-style employers generate especially diffuse candidate pools.

6.2 B. Preparing Revelio profiles for matching

The Revelio side is pre-processed to focus on workers who could plausibly belong to the recent H-1B applicant pool. The cleaned profile file keeps users with U.S.-based positions at employers in the linked employer sample and uses position histories beginning in 2015 or later. Open-ended jobs are cleaned conservatively, and positions are collapsed to the employer level around the lottery date.

The pipeline also removes likely duplicate profiles. A pair of profiles is treated as a duplicate only when the profiles share a normalized name and the same employer, and when at least one additional overlap condition is satisfied: overlapping job dates, overlapping education dates, or a common school. Within such duplicate sets, the profile with the richer work history is retained. This is designed to avoid collapsing genuinely different workers who happen to share a common name at a large employer.

Finally, the Revelio cleaning stage excludes profiles with clear evidence of U.S. secondary schooling, which are less likely to belong to the foreign-born population of interest. It also records whether the profile shows U.S. postsecondary education and whether the worker appears to qualify for the advanced-degree exemption. The paper's preferred linked sample later requires confirmed U.S. postsecondary education, but that restriction is imposed after the broader cleaned profile file is built.

6.3 C. Variables inferred on the Revelio side

Because the H-1B data are deidentified, the person-level match relies on a small set of fields that are either observed or inferable in both sources.

Country of origin. Country evidence is built from several sources. School histories identify the country of attended institutions. Name-based models provide both country-level and broader regional predictions. Position histories provide an additional check when a user's observed work history lines up with the same origin country. The pipeline combines these inputs into a country-level signal and a broader regional signal, while separately flagging cases in which the country prediction is unusually uncertain.

Year of birth. Estimated year of birth is inferred from education timing. When a high school record is available, the birth-year estimate comes from the implied age at high school completion. Otherwise it is backed out from postsecondary start and end dates, with adjustments for degree type.

Sex. The administrative records contain sex, while Revelio does not. The code therefore constructs a name-based sex probability using more than one name model and treats this as a noisy comparison field rather than a deterministic identifier.

Education and occupation. Education histories are also used to identify U.S. postsecondary attendance, advanced-degree status, the year of the most recent U.S. graduation, and broad field of study. Position titles are cleaned and mapped to H-1B-relevant occupation rankings. These fields are useful both for sharpening the match and for defining heterogeneity in the later analysis.

6.4 D. Building applicant-specific candidate pools

For each H-1B application, the candidate pool is defined inside the linked employer-year cell. In the current implementation, the first pass is deliberately broader than exact country matching: applicants are initially paired with Revelio users at the same employer who fall in the same broad region of origin. Exact country agreement then receives much more weight at the ranking stage. This design allows the pipeline to retain plausible cases when country classification is noisy without treating broad regional agreement as equivalent to a true country match.

Timing restrictions are central. A candidate must appear to have started at the sponsoring employer within the few years before the lottery and still be employed there at the March filing date associated with that lottery. In the code, the reference date is March 1 of the calendar year preceding the fiscal-year label in the H-1B data. Candidates are also screened using age and education timing so that the implied life cycle is consistent with a recent student-to-work transition rather than with a much older worker. Profiles that clearly look unrelated to the skilled H-1B applicant pool are removed as well.

These restrictions reflect the economics of the modern H-1B system. For many recent applicants, the relevant comparison is not between a worker outside the United States and a worker already employed at the sponsor, but between lottery winners and lottery losers who are both already working for that employer before the visa decision is realized. That institutional fact is what makes the person-level linkage tractable.

6.5 E. Ranking candidates and carrying forward uncertainty

After the candidate pool is defined, each applicant-candidate pair receives a composite score. Country-of-origin agreement is the dominant component of the score. Consistency in sex and year of birth provides secondary information. Occupation provides an additional refinement but does not drive the match on its own. Candidate scores are also adjusted for employer-link quality: a candidate receives more credit when the candidate's own position history directly supports the sponsoring employer match and less credit when the employer crosswalk itself is weak.

The score is then normalized within applicant to form match weights that sum to one across retained candidates. Those weights are the object carried into the regression analysis. This is important conceptually. A deterministic one-to-one match would hide uncertainty and make a single mistaken assignment disproportionately influential. By contrast, the weighted approach preserves ambiguity transparently and turns linkage noise mainly into attenuation rather than into arbitrary reclassification.

The code records several diagnostics that are used later in the paper: the number of retained candidates per applicant, the top candidate's weight, the gap between the top two candidates, and indicators for ambiguous cases in which the top candidate is only weakly

separated from the runner-up.

Table 6: How the current scoring rules are used

Component	Role in the linkage
Country of origin	Main identifying signal. Exact country agreement matters most, while broader regional agreement is used only as a fallback when country evidence is weak or noisy.
Sex and year of birth	Secondary signals that down-rank otherwise plausible candidates when the profile does not fit the administrative record.
Employer-link quality	Candidate scores are discounted when the employer crosswalk is weak and receive full credit when the candidate's own job history clearly supports the sponsoring employer match.
Occupation and field	Low-weight refinements that help favor profiles consistent with skilled H-1B-type work.
Ambiguity flag	Applications are flagged when the leading candidate is not clearly separated from the runner-up.
Strict sample	Keeps only dominant top-ranked matches with strong absolute score, strong employer-link evidence, strong country evidence, and a usable birth-year estimate.

6.6 F. Linked analysis samples

The pipeline produces several linked samples for analysis.

- **Broad linked sample.** Keeps all retained candidates after the main plausibility filters.
- **Preferred U.S.-educated sample.** Further requires the matched Revelio profile to show U.S. postsecondary education. This is the paper's preferred sample because it most closely targets workers likely to have entered the sponsoring firm through OPT.

- **Multiplicity-capped samples.** Restrict attention to applications with only a small number of retained candidates, which reduces attenuation from diffuse match weights.
- **Pre-filtered sample.** Excludes origin-country groups that generate especially large candidate pools in Revelio, mainly because both H-1B application counts and online-profile coverage are extremely high.
- **Strict sample.** Keeps only the top-ranked candidate when that candidate dominates the applicant's weight, clears a high absolute score threshold, has strong employer-link evidence, and has strong country evidence.

These variants are useful for a simple reason: if linkage error mainly attenuates treatment effects, estimates should generally become larger in higher-precision samples even as the sample size falls.

6.7 G. Outcomes constructed from linked profiles

Once workers are linked, I construct outcomes from Revelio position and education histories and aggregate them to the applicant level using the match weights. The retention outcome is defined at the employer level rather than the job-title level, which makes it less sensitive to noisy title coding and internal reorganizations. Likewise, the geographic outcomes are based on observed location fields, so they should be interpreted as evidence on observed geographic attachment rather than as direct measures of legal status.

Table 7: Outcome concepts used after the linkage

Outcome	Construction in words
Still at sponsoring firm	Whether the linked worker is observed in an active job at the sponsoring employer one, two, or three years after the lottery.
Move to a different employer	Whether the worker starts a post-lottery job at a different employer.
Internal move or promotion	Whether the worker starts a new post-lottery position at the same employer.
Observed in the United States	Whether any job or education record places the worker in the United States in the relevant post-lottery year.
Observed in origin country	Whether any job or education record places the worker in the worker's country of origin in the relevant post-lottery year.
New education	Whether the worker begins a new education spell after the lottery.
Compensation	An annual compensation measure built from Revelio's imputed pay fields and the share of the year covered by each active job.

6.8 H. Validation

The main ex-post validation exercise uses lottery winners, for whom the administrative files contain much richer petition information than for lottery losers. For those cases, I compare the linked Revelio profiles to the administrative records on education level, advanced-degree institution when relevant, field of study, and job title. Match quality rises strongly with the linkage weight and is highest in the strict sample, which supports the interpretation of the weights as meaningful measures of matching confidence rather than arbitrary scaling factors.

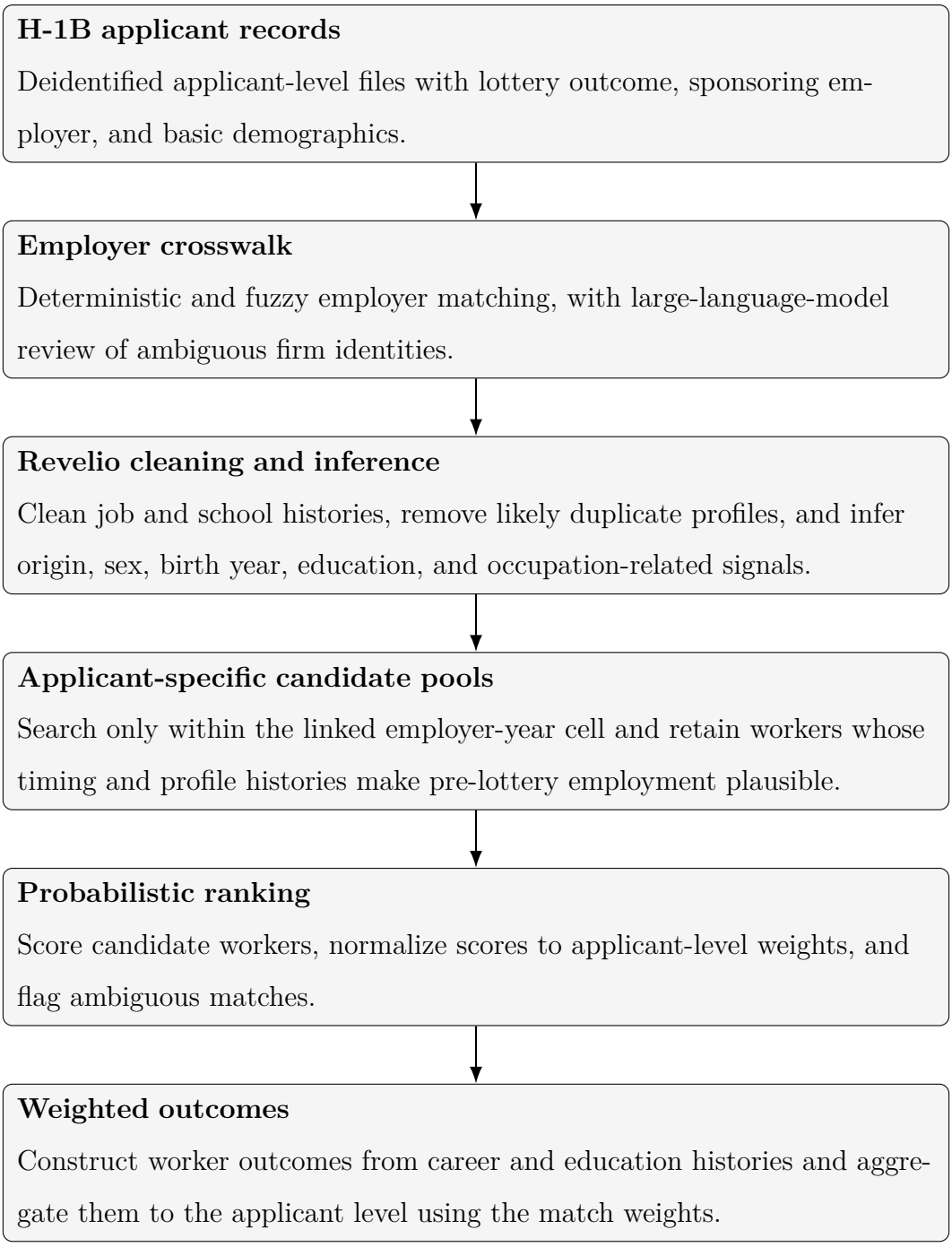

Figure 2: Linkage pipeline from deidentified H-1B records to applicant-level outcomes